

# MISUN KIM

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## RESEARCH INTERESTS

I study the reliability, safety, and controllability of interventions in foundation models, focusing on when similar internal representations in multilingual LLMs lead to different behavioural responses under cross-lingual and cross-task transfer.

## EDUCATION

### MSc Speech and Natural Language Processing — University of Sheffield

Sheffield, UK | Sep 2025 - Sep 2026 | Predicted: Distinction

Dissertation: Cross-lingual generalisation of representation-level interventions in multilingual LLMs.

### BE in Industrial Management Engineering — Hankuk University of Foreign Studies

South Korea | Mar 2019 - Feb 2024 | Double Major: Central Asian Studies | GPA 3.94/4.5 | First Distinction Scholarship (2023)

## RESEARCH EXPERIENCE

### MSc Dissertation Research — Multilingual LLMs

University of Sheffield | Supervisor: Dr Cass Zhao | Feb 2026 - Present

- Investigated the robustness of activation-level interventions in multilingual LLMs using Contrastive Activation Addition (CAA), testing whether English-derived steering vectors transfer consistently across three model families and five languages.
- Found that high cross-lingual representational similarity did not reliably predict behavioural transfer, revealing a gap between geometric alignment and intervention robustness.
- Designed layer-wise evaluations with bootstrap confidence intervals, held-out sets, and Gaussian and permutation random-vector baselines to characterise failure modes and distinguish systematic effects from noise.
- Extended analysis to TruthfulQA and cross-task settings, finding that interventions targeting one behaviour can also shift performance on another task and motivating mechanistic analysis of off-target effects.

### Undergraduate Researcher — ICT Mentoring Project, Ministry of Oceans and Fisheries

Supervisor: Gyu-Young Lee (KAIST) | Mar 2023 - Nov 2023

- First-authored an LSTM study on container freight forecasting across five universities, reducing RMSE by 71% over the baseline (published ACK 2023).

## KEY PROJECTS

### Multilingual Post-Training Safety Evaluation Framework

Aug 2026 - Present

- Built a reproducible multilingual LLM safety-evaluation pipeline for SFT, DPO, and RLVR checkpoints, combining typed experiment/result schemas, benchmark-native scoring, deterministic parallel-sample validation, and automated GitHub Actions CI.
- Ran a 50-item parallel EN/KO pilot on OLMo 3 7B using A40 GPUs, finding a consistent harmful-request refusal gap between Korean and English across all three post-training stages, motivating larger-scale evaluation of cross-lingual safety consistency.

### ESG Compliance Checker

Mar 2026 - Present

- Built an end-to-end compliance verification system over 11 corporate ESG reports, mapping regulatory criteria to retrieved evidence and producing traceable covered/partial/missing judgements using hybrid retrieval and LLM-assisted verification.
- Developed and audited a manually annotated retrieval gold set; hybrid retrieval with reranking improved Hit@5 from 83.3% to 100% and MRR from 0.628 to 0.819 over BM25-only retrieval.
- Used retrieval and verifier error analysis to identify annotation gaps and evidence-selection failures, then modularised the system with FastAPI, automated testing, Docker, and a separate Streamlit interface.

## WORK EXPERIENCE

### Sustainability Research Analyst — Eon Consulting Group

South Korea | Mar 2025 - May 2025

- Contributed to 170+ page sustainability reports for major public and financial organisations, conducting TNFD climate scenario analysis and developing a CSDDD-aligned supply-chain risk assessment framework.

### Field Service Manager (Data & Operations) — Payhere

South Korea | Mar 2024 - Aug 2024

- Built workflow automation using the Slack API, Google Sheets, and Looker Studio to analyse 10,000+ service records, reducing unnecessary field visits by 17%.

## TEACHING EXPERIENCE

### Language Tutor (Private)

Mar 2021 - Nov 2025

- Provided individualised Korean/English instruction to 5+ learners aged 6-18, from early literacy to CSAT-level advanced reading.

## SKILLS

**Programming:** Python, SQL

**ML & NLP:** PyTorch, Hugging Face Transformers, scikit-learn, clustering (k-means, hierarchical)

**LLM Interpretability & Safety:** activation steering (CAA), representation analysis, layer-wise interventions, cross-lingual behavioural evaluation, post-training safety evaluation, refusal analysis, experimental controls, LLM-as-judge

**Research Infrastructure:** HPC/SLURM, Weights & Biases, Git, Docker, FastAPI, Flask, ChromaDB

## LANGUAGES

Korean (Native), English (Fluent), Spanish (Intermediate), Kazakh (Intermediate)

## PUBLICATIONS & MANUSCRIPTS

Kim, M., Kim, Y., Kim, E., Lee, B., Han, Y., & Lee, G. (2023). *Implementation of Container Volume Prediction Technology using Deep Learning*. ACK Conference Proceedings, 30(2).

Manuscript in preparation for short paper submission (2026). *Cross-lingual and cross-task generalisation of representation-level interventions in multilingual LLMs*.